

String Manipulation Programs

1. Counting Characters

Write a program to count the number of occurrences of the character 'e' or 'E' in the given string:

Input: s = "javaEEdeve"

Expected Output: e/E count: 3

2. Finding Vowels

Write a program to find lowercase vowels from the given string and display both the vowels and their count.

Input: s = "javadev"

Expected Output:

- Vowels: a, a, e
- Count: 3

3. Reversing a String

Write a program to reverse a string.

Input: s = "javadev"

Expected Output: vedavaj

4. Checking for Palindrome

Write a program to check whether a given string is a palindrome.

Input:

a. "madam"

b. "Mom"

Expected Output:

- "madam" is a palindrome.
- "Mom" is a palindrome.

5. Counting Words in a String

Write a program to count the number of words in a string.

Input: s = "I am a java developer"

Expected Output: 5

6. Printing Unique Characters

Write a program to print all unique characters in a string.

Input: s = "javajavajavadevdevdev"

Expected Output: javde

7. Replacing Characters in a String

Write a program to replace all occurrences of 'e' with 'a' in the given string.

Input: s = "java development"

Expected Output: "java davalopmant"

8. Replacing Substrings

Write a program to replace "java" with "core java" in the given string.

Input: s = "java development"

Expected Output: "core java development"

9. Removing Spaces

Write a program to remove all spaces from a given string.

Input: s = "java development"

Expected Output: "javadevelopment"

10. Removing Capital Letters

Write a program to remove all uppercase letters from a string.

Input: s = "jAvA DeVeloPer"

Expected Output: "jveloer"

11. Removing Small Letters

Write a program to remove all lowercase letters from a string.

Input: s = "jAvA DeVeloPer"

Expected Output: "ADA DVP"

12. Removing Digits

Write a program to remove all digits from a string.

Input: s = "ja123vaDEveloper"

Expected Output: "javaDEveloper"

13. Removing Vowels

Write a program to remove all vowels from a string.

Input: s = "ja123vaDEveloper"

Expected Output: "j123vDvlpr"

14. Counting Substring Frequency

Write a program to count the frequency of a given substring in a string.

Input:

String str1 = "we work to live and live to be happy live";

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String word1 = "live";
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Expected Output: 3

15. Finding the Longest Word

Write a program to find the longest word in a string.

Input: s = "I am a java developer"

Expected Output: "developer"

16. Counting Character Frequency

Write a program to count the frequency of each character in a given string.

Input: s = "javadev"

Expected Output:

A - 2

D - 1

E - 1

J - 1

V - 2

17. Reversing Words in a Sentence

Write a program to reverse the order of words in a sentence.

Input: s = "I love Java"

Expected Output: "Java love I"

18. Valid Anagram

Check if two strings are anagrams

Test Cases:

Input: s = "anagram", t = "nagaram"

Output: true

Input: s = "rat", t = "car"

Output: false

19. Longest Substring Without Repeating Characters

Test Cases:

Input: "abcabcbb"

Output: 3

Input: "bbbbbb"

Output: 1

Input: "pwwkew"

Output: 3

20. Valid Parentheses

Test Cases:

Input: "()"

Output: true

Input: "()[]{}"

Output: true

Input: "("

Output: false

Input: "([)]"

Output: false

21. Longest Common Prefix

Test Cases:

Input: ["flower", "flow", "flight"]

Output: "fl"

Input: ["dog", "racecar", "car"]

Output: ""

22. Decode String

Test Cases:

Input: "3[a]2[bc]"

Output: "aaabcbcb"

Input: "3[a2[c]]"

Output: "accaccacc"

Input: "2[abc]3[cd]ef"

Output: "abcbccdcddcdef"

23.Isomorphic Strings

Test Cases:

Input: s = "egg", t = "add"

Output: true

Input: s = "foo", t = "bar"

Output: false

Input: s = "paper", t = "title"

Output: true

24.Convert Number to Binary

Test Cases

Input: n = 5

Output: "101"

Input: n = 10

Output: "1010"

Input: n = 0

Output: "0"

Input: n = 1

Output: "1"

